Notes: Brandsaas and Kvaerner (JFE, 2026)

Partial Homeownership: A Quantitative Analysis

Contents

1 Big picture 3

2 Data and measurement (key objects) 3

2.1 Institutional setting and observed PO contract . . . 3

2.2 Observed PO-user facts . . . 4

2.3 Observed household panel and calibration targets . . . 4

2.4 Unobservables, timing, and empirical applications . . . 4

3 Models and empirical specifications (all equations + interpretation) 5

3.1 Preferences and ownership utility . . . 5

3.2 Labor income process . . . 5

3.3 House prices, rents, and housing sizes . . . 6

3.4 Wealth, mortgages, and borrowing constraints . . . 6

3.5 The PO contract inside the model . . . 6

3.6 An age-dependent PO participation cost . . . 7

3.7 Budget equations and law of motion . . . 7

3.8 Recursive problems and constraints . . . 7

3.9 Calibration and estimation strategy . . . 8

3.10 WTP and default calculations . . . 9

4 Identification and instruments (what makes the estimates credible) 9

4.1 What is *not* doing the work . . . 9

4.2 How the baseline model is identified . . . 9

4.3 How the ownership elasticity is identified . . . 9

4.4 Why the contract data are essential . . . 10

4.5 Additional credibility checks . . . 10

4.6 Relation to the literature 10

5 Tables and figures 11

5.1 Figure 1: Summary statistics from PO contracts 11

5.2 Figure 2, Table 1, and Figure 3: Calibration, parameters, and model fit 11

5.3 Figure 4, Table 2, Figure 5, and Figure 6: Ownership elasticity and policy functions 12

5.4 Figure 7: Aggregate housing outcomes after PO introduction 12

5.5 Figure 8, Figure 9, and appendix sensitivity tables: Willingness-to-pay for PO 13

5.6 Figure 10, Figure 11, and Figure 12: Debt-to-income effects 13

5.7 Table 3: Default rates with and without PO 14

5.8 Appendix robustness: option value, housing supply, and risk aversion 14

6 Short summary 14

7 Conclusion 14

7.1 Core conclusion (what the paper establishes) 14

7.2 Economic intuition 15

7.3 Takeaway 15

1 Big picture

- **Question.** What happens if households can buy only a *fraction* of a home and rent the rest? Can partial ownership (PO) relax borrowing constraints, raise welfare, and still be compatible with financial stability?
- **Setting.** The paper studies a real for-profit Norwegian PO contract offered by OBOS and embeds it in a standard life-cycle housing model with renting, traditional homeownership, stochastic income, and stochastic house prices.
- **Core challenge.** Housing is lumpy and heavily regulated by mortgage constraints. Many young or low-wealth households would like some exposure to homeownership but cannot afford the full fixed investment. A useful model must match both the institutional details of the contract and the life-cycle patterns of wealth and ownership.
- **Main modeling innovation.** The paper augments a standard life-cycle model with a contract that allows households to choose an ownership share between 50% and 90%, and it estimates a new preference parameter—the **ownership elasticity**—that governs how much of the utility premium from ownership is obtained at partial shares.
- **Main results.** PO increases the number of households who own *some* housing, especially among the young. The mean willingness-to-pay (WTP) for access to PO is economically large, around 5%–7% of after-tax income for households aged 25–45. PO raises the average debt-to-income ratio because many renters become borrowers, but it compresses the right tail of debt and lowers default risk, especially after house-price declines.
- **Key empirical message.** PO is not just a curiosity. In the model, it shifts households from the extensive margin of *no ownership* into *partial ownership*, while reducing the number of households that lever up all the way into traditional ownership.
- **Why this paper matters.** It provides a structural framework for studying a fast-growing housing-finance innovation. More broadly, it shows how changing the *ownership form* of housing can affect affordability, welfare, leverage, and stability simultaneously.

2 Data and measurement (key objects)

2.1 Institutional setting and observed PO contract

- **Country setting.** Norway is a high-homeownership economy: about 80% of households own and 20% rent. The rental market is relatively lightly regulated, while owner-occupied housing receives favorable tax treatment.
- **Mortgage regulation.** The key borrowing constraints are a loan-to-value limit of 0.85 and a debt-to-income limit of 5.0. Borrowers must also be able to withstand a mortgage-rate increase of several percentage points.
- **Observed PO contract.** The paper uses the most common for-profit PO contract ("deleie") offered by OBOS. Households must buy at least 50% of the unit and can later raise ownership in 10 percentage point increments. They pay rent on the part not owned, are responsible for in-unit maintenance, and share common costs according to ownership share.

- **Pricing rule for increasing ownership.** When a household buys a larger share later, the relevant price is the maximum of the initial contract price and the current market-indexed price. This embeds an option-value element in the contract.
- **Contract horizon.** OBOS guarantees the contract for ten years, after which it can list the property for sale. The paper models the main contract terms but omits this put option in the baseline for tractability.
- **Survey evidence on demand.** A survey in Oslo reports that 37% of all households and 70% of renters would consider PO in their next housing decision.

2.2 Observed PO-user facts

- **Main PO data.** The paper uses proprietary contract-level data from OBOS, the largest homebuilder in Scandinavia and the largest PO supplier in Norway.
- **Buyer composition.** The average PO buyer is about 35 years old, which strongly suggests that the contract is aimed at first-time or relatively young buyers rather than retirees.
- **Initial ownership choice.** About 59% of buyers start at the minimum 50% ownership share, and 31% of partial owners have increased their share at least once.
- **Housing units.** The typical unit is a small apartment in an urban multifamily building: on average 2.67 rooms, about 59 square meters, and located in a building with many units.
- **PO take-up in the market.** The share of new-unit sales using PO has risen quickly in the markets where it is offered, and builders report that legal restrictions on the share of PO units per building constrain supply.

2.3 Observed household panel and calibration targets

- **Administrative data.** Wealth, homeownership, income, and education come from the Norwegian Tax Registry and Statistics Norway.
- **Income measure.** Household income is broadly defined as gross salary income, pension income, net capital income, and government transfers, net of taxes.
- **Housing-size data.** The distribution of unit sizes comes from Eiendomsverdi AS.
- **Calibration targets before PO.** The key targeted moments in the pre-PO model are average net worth and homeownership rates by age (25–50).
- **Calibration target with PO.** The key targeted moment for the new PO-specific parameter is the average initial ownership share among new PO users, which is 57% in the data.

2.4 Unobservables, timing, and empirical applications

- **Unobservables.** Households differ in wealth, education, permanent income shocks, transitory income shocks, and the realized path of house prices.
- **Timing.** House-price and income shocks are realized at the start of the period. Households then choose consumption, housing, ownership share, and liquid wealth for the next period.

- **Baseline versus extension.** The baseline model does not allow mortgage default because default rates in Norway are very low. Default is introduced later as an extension to study stress scenarios.

- **Empirical applications.** The paper uses the model to study four objects: take-up rates and the housing ladder, willingness-to-pay for PO, debt-to-income ratios, and mortgage default under stress.

3 Models and empirical specifications (all equations + interpretation)

3.1 Preferences and ownership utility

Households choose consumption, housing services, and ownership status over the life cycle:

$$\max_{\{C_a,H_a,S_a\}} \mathbb{E}_{24}\left[\sum_{a=24}^{T}\beta^{a-24}\frac{\left(C_a^{1-\eta}H_a^{\eta}\chi(S_a)\right)^{1-\gamma}}{1-\gamma}\right]+\beta^{T-23}B(Q). \tag{1}$$

The key new object is the ownership premium

$$\chi(S)=1+\chi S^{\alpha},$$

where $S=0$ for renters, $S=1$ for traditional owners, and $S\in(0,1)$ for partial owners.

- **Interpretation.** The paper allows the utility gain from ownership to be only partially proportional to ownership share. This is the core novelty. A household may obtain most of the nonfinancial benefits of owning even when it owns only part of the unit.

- **Role of α.** If α is very small, even a low ownership share delivers most of the ownership premium. If α is very large, only near-full ownership looks like true ownership in utility terms.

- **Bequest.** The terminal bequest function converts end-of-life wealth into an annuity flow and evaluates utility from the implied consumption and housing bundle. The exact formula matters less than the economic role: it keeps households from fully running down wealth late in life.

3.2 Labor income process

Before retirement, income is

$$\ln(Y_{a,e})=f(a,e)+\nu_{a,e}+\epsilon_{a,e}, \tag{3}$$

with persistent component

$$\nu_{e,a}=\nu_{e,a-1}+u_{e,a}, \tag{4}$$

and education-specific shock variances. After retirement, income becomes a fixed fraction of income at retirement age.

- **Interpretation.** Labor income has a deterministic life-cycle profile by education plus permanent and transitory shocks.

- **Why it matters.** Borrowing constraints make the joint distribution of income and wealth central for tenure choice. PO is especially valuable when adverse income realizations make full ownership hard but some ownership still desirable.

3.3 House prices, rents, and housing sizes

House prices follow a random walk with drift:

$$\Delta p_t^H = \mu + \sigma_h Z_t, \tag{5}$$

where $p_t^H = \ln(P_t^H)$ and $Z_t \sim N(0,1)$. The rent-to-price ratio κ is constant.

The model contains four discrete housing sizes. The smallest is rent-only, the largest is own-only, and the two middle sizes can be rented, owned traditionally, or purchased through PO.

- **Interpretation.** The paper uses a segmented housing market that lets PO compete in the realistic middle of the housing-size distribution rather than everywhere.
- **Why prices matter.** PO is valuable partly because it reduces exposure to house-price risk. That mechanism only appears in a model with explicit house-price uncertainty.

3.4 Wealth, mortgages, and borrowing constraints

Liquid wealth earns a piecewise return:

$$r(LW) = \begin{cases} r_f + \theta, & LW < 0, \\ r_f, & \text{otherwise.} \end{cases} \tag{6}$$

Borrowing is constrained by both loan-to-value and debt-to-income limits.

- **Interpretation.** Mortgages are one-period loans rolled over each year, so the household's net cash position is summarized by liquid wealth.
- **Why this matters.** PO changes the size of the initial mortgage needed to get onto the housing ladder. This is the channel through which it relaxes financing constraints.

3.5 The PO contract inside the model

The ownership grid is

$$S' \in \{0.0, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}. \tag{14}$$

Partial owners may raise ownership shares later, but they cannot reduce them without selling the home and re-entering under a new contract.

The cost of increasing the ownership share from S_a to S_{a+1} is

$$\bar{P}_t H_a \Delta S_{a+1}(1 + m_b) + l_c, \qquad \bar{P}_t = \max\{P_t^H, P_0^H\}. \tag{7}$$

- **Interpretation.** The household pays the higher of the initial and current price when increasing its share. This mimics the contract's embedded option-value feature.

- **Recurring PO costs.** Partial owners pay market rent on the non-owned share, pay all interior maintenance, and pay common-property costs in proportion to ownership.
- **Important finding.** The option-value element turns out not to be quantitatively central because households can often "reset" the contract by selling and re-entering if prices fall enough.

3.6 An age-dependent PO participation cost

To prevent the model from implying unrealistically high PO take-up among retirees, the paper adds

$$c(a) = \max\{0, a - K\}. \tag{8}$$

- **Interpretation.** This is a reduced-form age cost that proxies for contract features and late-life considerations that make PO unattractive for seniors.
- **Why this matters.** Without it, PO behaves too much like a reverse mortgage in retirement.
- **Caveat.** This is an explicitly ad hoc modeling choice. The paper is transparent about this, and the assumption is best viewed as a tractable way to keep the model aligned with the observed near-absence of seniors in the PO market.

3.7 Budget equations and law of motion

For renters and traditional owners, the budget equation is

$$W + Y = C + LW' + ac(P, S, H, S', H') + (1 - S')\kappa PH' + S'PH'. \tag{9}$$

For a partial owner, changing the ownership share adds an extra term:

$$W + Y = C + LW' + ac(P, S, H, S', H') + (1 - S')\kappa PH' + S'PH' + \mathbf{1}_{\Delta S' \neq 0,\, \Delta H'=0}\, \Delta S'\, \bar{P}_t H'. \tag{10}$$

Next period wealth is

$$W' = LW'(1 + r(LW')) + \mathbf{1}_{S'>0} P'H'\big(S'(1 - \delta) - \tau\big). \tag{11}$$

- **Interpretation.** The household allocates current resources across current consumption, next-period liquid wealth, and the housing choice for the next period.
- **Economic role of PO.** Relative to full ownership, PO lowers the required equity and mortgage needed to access the owned housing stock, but it does not make housing free: the household still pays rent on the share not owned.

3.8 Recursive problems and constraints

For renters and traditional owners, the recursive problem is

$$V(\Xi) = \max_{C, H', LW', S'} \big\{u(C, H') + \beta \mathbb{E}\big[V(\Xi')\big]\big\} \tag{12}$$

subject to

$$C > 0, \tag{13}$$

$$H' \in \mathcal{H}(S'), \tag{15}$$

$$LW' \geq -LTV \times PH'S', \tag{16}$$

$$LW' \geq -DTI \times YS'. \tag{17}$$

For partial owners, the initial contract price P_0^H becomes an extra state variable:

$$V(\Xi; P_0^H) = \max_{C,H',LW',S'} \left\{ u(C, H') + \beta \mathbb{E}\left[V(\Xi'; P_0^H)\right] \right\}. \tag{18}$$

- **Interpretation.** The state vector for partial owners is richer because past contract terms matter for future share adjustments.
- **Policy-function logic.** Without PO, renters and owners face threshold rules: when wealth is high enough, they jump discretely into homeownership. With PO, this jump is replaced by a staircase of gradually rising ownership shares.

3.9 Calibration and estimation strategy

The parameterization has three stages.

- **Stage 1: external calibration.** The paper takes from data, institutional details, or prior literature parameters such as $\gamma = 2.0$, $\eta = 0.3$, $r_f = 0.0143$, mortgage premium $\theta = 0.016$, $LTV = 0.85$, $DTI = 5.0$, price growth $\mu = 0.023$, and price volatility $\sigma_h = 0.0564$.
- **Stage 2: baseline SMM.** The discount factor β and ownership utility shifter χ are estimated by simulated method of moments to match age profiles of wealth and homeownership before PO is introduced.
- **Stage 3: PO-specific SMM.** The ownership elasticity α is estimated by matching the average initial ownership share among new PO users.

The SMM estimator for the baseline parameters is

$$\omega^*(\Omega) = \arg\min_{\omega} \left[\hat{m}(\omega) - \hat{m}\right]' \Omega \left[\hat{m}(\omega) - \hat{m}\right], \qquad \omega \equiv \{\beta, \chi\}. \tag{19}$$

The internally estimated parameters are

$$\beta = 0.944, \qquad \chi = 0.061, \qquad \alpha = 0.272.$$

- **Interpretation.** The ownership-share elasticity is estimated after the rest of the housing model is pinned down. This avoids contaminating pre-PO homeownership moments with a parameter that only matters when PO exists.
- **Economic meaning of $\alpha = 0.272$.** A household owning 50% of the home receives about $0.5^{0.272} \approx 80\%$ of the homeownership utility premium.

3.10 WTP and default calculations

The one-time willingness-to-pay for access to PO is defined as

$$\mathrm{WTP}(\Xi) = \left\{ c \in \mathbb{R} : \; V(\Xi) = \tilde{V}(W - c, H, S, \nu, P, a) \right\}, \tag{20}$$

where $\tilde{V}$ is the value function in the economy with PO.

In the default extension, households receive a default flag $D \in \{0, 1\}$, cannot borrow while in default, and re-enter with probability λ. The default rule is

$$D'(\Xi) = \Big(V(W, S, H, \nu, P, a; D = 0) < V(W = 0, S = 0, H = 1, \nu, P, a; D = 1) + \zeta \Big). \tag{21}$$

The paper calibrates

$$\lambda = 0.75, \qquad \zeta = -0.018.$$

- **Interpretation.** WTP translates utility gains from PO into a money metric. The default extension translates changes in leverage and ownership share into a financial-stability object.
- **Calibration logic.** The default cost is chosen to match the observed forced-sale rate, while the removal probability is set to generate a short expected exclusion from credit markets.

4 Identification and instruments (what makes the estimates credible)

4.1 What is *not* doing the work

- This is **not** a reduced-form policy experiment or a quasi-experimental rollout design.
- The paper does **not** identify PO's effects from cross-regional or cross-cohort exposure variation.
- There are therefore no external instruments in the usual IV sense. Identification is structural and comes from the mapping between parameters and moments.

4.2 How the baseline model is identified

- The discount factor β is identified from the life-cycle accumulation of wealth.
- The ownership utility shifter χ is identified from homeownership rates over the life cycle.
- The paper uses 52 age-specific moments (wealth and homeownership at each age from 25 to 50) for just two parameters, so the baseline is highly overidentified.
- The estimated model then reproduces the observed age profiles reasonably well, which is the first major credibility check.

4.3 How the ownership elasticity is identified

- The new PO-specific parameter α is identified from the **average initial ownership share** of new PO buyers.

- The logic is sharp. If low ownership shares already provide nearly all the utility benefits of ownership, households choose the minimum 50% share. If they do not, households buy a higher share immediately.
- The data moment is informative because the contract itself only offers a discrete set of ownership shares and because 59% of buyers choose exactly the minimum 50% share.
- Matching an average initial share of 57% yields $\alpha = 0.272$, implying substantial utility from partial ownership but not full equivalence with outright ownership.

4.4 Why the contract data are essential

- The proprietary OBOS data are crucial because they reveal actual PO usage, buyer age, and chosen ownership shares.
- Without these data, the paper could not separately identify the new PO-specific preference parameter.
- The real-world contract also disciplines the model through the minimum share rule, adjustment fees, maintenance split, and pricing of share increases.

4.5 Additional credibility checks

- **Model fit.** The pre-PO model matches wealth and homeownership profiles reasonably well.
- **PO-target moment fit.** The model hits the average initial ownership share exactly by construction and is close on the share of buyers choosing 50%.
- **Sensitivity checks.** The appendix shows that the main results are robust to changes in the size of PO-eligible housing units, the option-value term in pricing, the degree of housing-supply elasticity, and risk aversion.
- **Honest limitation.** The age-dependent PO cost is a reduced-form patch rather than a primitive friction. It helps the model fit the age composition of PO users, but it also means the model is not a fully structural explanation of why retirees avoid PO.

4.6 Relation to the literature

- The paper sits in the life-cycle housing literature following Cocco (2005), Attanasio et al. (2012), and Davis and Van Nieuwerburgh (2015).
- It is related to work on alternative housing-finance products such as reverse mortgages and shared-equity/shared-appreciation contracts.
- Its main contribution is to bring a *for-profit* partial ownership contract into a quantitatively disciplined housing model and connect it to affordability and stability outcomes.

5 Tables and figures

5.1 Figure 1: Summary statistics from PO contracts

Read:

- PO is not a niche contract in the raw provider data: its sales share rises quickly wherever it is offered.
- The typical PO buyer is young (around age 35), which is exactly the demographic one would expect to be most constrained by downpayment and DTI rules.
- About 59% of buyers start at the minimum 50% share, which immediately tells you that households value “some ownership” a lot.
- About 31% later increase their ownership share, which supports the paper’s interpretation of PO as a stepping stone toward fuller ownership rather than a dead-end tenure form.

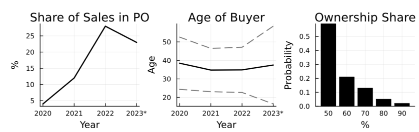
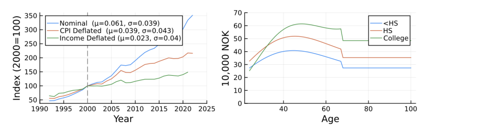


Fig. 1. Summary Statistics from PO Contracts. The left panel plots the share of sales in PO, in regions where PO is offered, with data through February 2023. The middle panel plots the mean and standard deviation of the buyer’s age. The right panel plots the initial ownership share.
Source: OBOS.



(a) House Prices (b) Calibrated Income Process

Fig. 2. Calibration. The figure shows the price index for existing homes in Norway in nominal and real terms, as well as the mean and standard deviation of the log growth rate.

5.2 Figure 2, Table 1, and Figure 3: Calibration, parameters, and model fit

Table 1
Model Parameters Superscript† denotes variables in NOK10,000. SN refer to publicly available tables produced by Statistics Norway.

Panel A: Externally Calibrated Parameters			
Parameter		Value	Source
Risk aversion	γ	2.0	Campbell and Cocco (2015)
Exp. share housing	η	0.3	Yao et al. (2015)
Annuity duration	L	3	Vestman (2019)
Risk-free rate	r^f	0.0143	Fagereng et al. (2017)
Mortgage premium	θ	0.016	SN 08175
Wage profiles	$f(a,e)$	Fig. 2(b)†	Data
Persistent income shock	σ_u	Tab. I.2	Data
Transitory shock	σ_ϵ	Tab. I.2	Data
Retirement income drop	ϕ_{ret}	0.842	Fagereng et al. (2017)
Rent-to-price ratio	κ	0.044	SN 06035, 09895
Sales cost	m_s	0.025	Commission and fees
Purchase cost	m_b	0.025	Tax code
Property maintenance	δ	0.02	Yao et al. (2015)
In-unit maintenance	τ	0.005	Yao et al. (2015)
Loan-to-value	LTV	0.85	Law
Debt-to-income	DTI	5.0	Law
Price growth	μ	0.023	SN 04751, 03014, 07230
Price shock	σ_h	0.0564	SN 04751, 03014, 07230
Rental sizes	$\mathcal{H}(0)$	[1.0, 1.365, 1.6]	Own calculation
Owner-occupied sizes	$\mathcal{H}(1)$	[1.365, 1.6, 2.34]	Own calculation
PO sizes	$\mathcal{H}((0,1))$	[1.365, 1.6]	Own calculation
PO purchase cost	l_b	0.2128†	Contract
PO change cost	l_c	0.8701†	Contract
PO sales cost	l_s	0.0†	Contract
Starting age	n/a	24	
Retirement age	K	67	
Final age	T	100	
Consumption floor	$\underline{c}$	10†	Welfare system
Initial wealth distr.	n/a	Fig. D.9(a)†	Data
Initial price distr.	n/a	Fig. D.9(b)†	Data

Panel B: Internally Estimated Parameters			
Parameter		Value	Standard Error
Discount factor	β	0.944	0.0001
Homeownership utility	χ	0.061	0.0003
Ownership share elasticity	α	0.272	0.0438

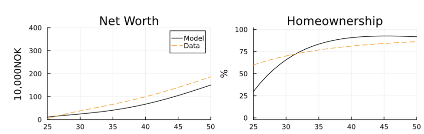


Fig. 3. Model Fit. The figure compares the predicted average net worth and the homeownership rate of households aged between 30 and 50 years using the parameter vector that solves Eq. (19) with the empirical counterparts.

Read:

- The calibration is rich and contract-specific: house-price drift and volatility, mortgage rules, transaction costs, rent-to-price ratio, and PO fees are all tied to data or institutions.
- The internally estimated parameters are economically reasonable: $\beta = 0.944$, $\chi = 0.061$, and $\alpha = 0.272$.

- Figure 3 shows that the baseline life-cycle model does a good job tracking net worth and homeownership across age before PO is introduced.
- This matters because the paper's PO counterfactuals rest on a sensible pre-PO benchmark rather than on an obviously misspecified housing model.

5.3 Figure 4, Table 2, Figure 5, and Figure 6: Ownership elasticity and policy functions

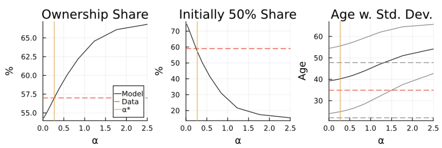


Fig. 4. Moments from Third-Stage Estimation. The horizontal dashed red line is the empirical moment, while the orange solid vertical line is the estimated parameter value for α. The gray lines in the right panel denote the mean plus/minus the standard deviation of age in the data (dashed) and the model (solid). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Model Fit - Third Stage. The ownership-elasticity α is set to match the average new initial ownership share (first row). *Source:* OBOS.

Moment	Model	Data	Target
Average new initial ownership share	57.0%	57%	Y
New owners owning 50%	57.3%	59%	N
Average Age	40.2	35.0	N
Std. Dev. Age	15.4	13.0	N

Read:

- Table 2 shows that the model exactly matches the key target: the average initial ownership share of 57% among new PO buyers.
- The estimated ownership elasticity implies that 50% ownership delivers roughly 80% of the utility premium from owning. That is why PO is attractive despite not giving full title.
- Figure 5 is the economic heart of the model: PO turns a discrete rent-versus-own threshold into a staircase of gradually increasing ownership shares.
- Figure 6 shows that α mainly determines *how much* of the home is initially purchased, whereas the discount factor and ownership premium shift the overall attractiveness of entering ownership.

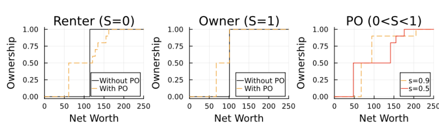
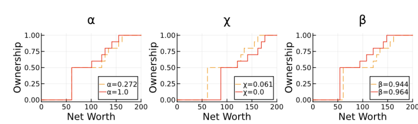


Fig. 5. Housing Choice Over the Wealth Distribution. The figures illustrate how young households decide to rent ($S = 0$), partially own ($0 < S < 1$), or own traditional ($S = 1$), as a function of wealth. The policy functions represent a 25-year-old high school graduate with median productivity who faces medium house prices. A wealth of 100 on the x-axis refers to 1 million NOK.



(a) Ownership-elasticity (b) Ownership Premium (c) Discount Factor

Fig. 6. Comparative Statics on Housing Decisions for Current Renters. The figures illustrate how young households decide to rent, partially own, or own as a function of wealth. The policy functions represent the same households as in Fig. 5.

5.4 Figure 7: Aggregate housing outcomes after PO introduction

Read:

- In the short run, PO sharply reduces renting among the young, with little immediate reduction in traditional ownership.
- In the long run, however, traditional ownership falls because many households use PO as a gradual path to ownership rather than jumping directly to 100% ownership.
- The paper's main quantitative summary is that about 30% of young households become partial owners in the long run.
- The average ownership share of the housing stock falls only slightly, so the big effect is on the *extensive margin* of entering ownership, not on a collapse in aggregate owned housing.

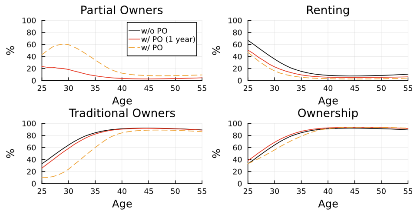


Fig. 7. Aggregate Outcomes After PO Introduction. The solid black line shows the outcomes without PO. The solid red line shows the outcome of interest one year after the introduction of PO. The orange dashed line shows the long-term outcomes with PO. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

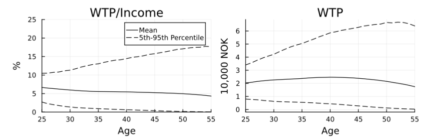


Fig. 8. Willingness to Pay (WTP) for PO. The figure plots the one-time WTP for PO as defined by Eq. (20). The left plot shows WTP as a percent of annual household income. The plot on the right shows WTP measured in 10,000 NOK.

5.5 Figure 8, Figure 9, and appendix sensitivity tables: Willingness-to-pay for PO

Read:

- Mean WTP for access to PO is large: for households aged 25–45, it is around 5%–7% of after-tax income.
- Renters value PO more than owners. The paper highlights that a 35-year-old renter is willing to pay about 10% of disposable income for access.
- WTP is highest for poorer, lower-wealth, and more constrained households—precisely the households excluded or made risky by standard mortgage rules.
- Sensitivity checks show that WTP rises when homeownership becomes more attractive, for example when expected house-price growth is higher, downpayment constraints are tighter, or rent-to-price ratios are lower.

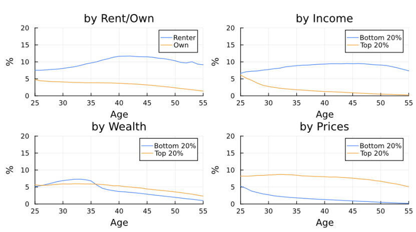


Fig. 9. Willingness to Pay (WTP) for PO by Household Type. The figure shows the average WTP for PO by household type. The *y*-axis shows WTP as a percent of after-tax income. For income, wealth, and house prices, we present the results for the top and bottom 20% of their distribution by age.

Table B.1
Housing Market Parameters and the Willingness-to-Pay for PO. The table reports the average WTP for PO, expressed as a percent of annual income, as we vary various housing-specific parameters. When we change total depreciation, we keep the shares allocated to the two types of maintenance constant. The first panel reports WTP by age groups, while the second panel reports averages for households (aged 25–55) within the top 20% of WTPs.

	Bench.	Price growth	Loan-to-value	Depreciation	Rent-to-price
New value		$\mu_h = 0.025$	$LTV = 0.8$	$\delta + \tau = 0.022$	$\kappa = 0.04$
Old value		$\mu_h = 0.023$	$LTV = 0.85$	$\delta + \tau = 0.025$	$\kappa = 0.044$
WTP, 25–55	5.5	6.7	6.3	6.9	5.7
Age 25	6.7	7.3	9.0	7.8	7.0
Age 35	5.5	6.8	6.4	6.9	5.7
Age 45	5.3	6.7	5.2	6.7	5.4
WTP, top 20%					
Age	38.5	39.8	35.7	39.2	38.5
Wealth	81.5	99.7	58.6	95.0	75.3
Own (%)	53.1	61.2	28.9	61.2	48.4
WTP (%)	12.8	15.2	14.5	15.9	12.8
<HS (%)	27.0	27.7	24.7	29.6	22.9

5.6 Figure 10, Figure 11, and Figure 12: Debt-to-income effects

Read:

- PO raises average DTI in the population because many renters take on mortgages to become partial owners.
- But conditional on borrowing, PO can reduce debt burdens because some households that would have stretched into full ownership instead choose smaller, safer loans through PO.
- Figure 11 isolates the key switchers: renters moving into PO drive the extensive-margin increase in debt, while owners and near-owners moving into PO show large DTI declines.
- Figure 12 shows the most important stability fact: PO compresses the right tail of the DTI distribution by reducing the number of households borrowing right up to the constraint.

5.7 Table 3: Default rates with and without PO

Read:

- Baseline default is tiny, but PO still lowers it: the all-household default rate falls from 0.06% to 0.02% in steady state.
- Under a house-price-drop stress test, PO approximately halves default: from 2.54% to 1.22% for the full sample.
- Under a negative income shock, the effect is smaller: from 0.24% to 0.16%. This is because PO users are often lower-wealth households who still struggle when income collapses.
- Under the combined house-price and income shock, default falls from 4.08% to 2.36%. This is one of the paper's strongest quantitative arguments that PO can improve stability even though it expands borrowing access.

5.8 Appendix robustness: option value, housing supply, and risk aversion

Read:

- The contract's option-value term matters little because households can sell and re-enter at lower prices when prices fall enough.
- Inelastic housing supply has only a modest effect on the main results; it raises PO use slightly but does not overturn the core welfare and stability conclusions.
- Higher risk aversion changes the composition of PO users and the level of WTP, but not the main qualitative findings that PO expands access to ownership and compresses risky leverage at the top of the DTI distribution.

6 Short summary

- **Conclusion.** Partial ownership works in the model as a financing technology that lets constrained households buy into housing gradually rather than all at once.
- **Identification insight.** The paper's main identification contribution is the estimation of the ownership elasticity from observed initial PO shares, conditional on a well-fitting pre-PO life-cycle housing model.
- **Main empirical lesson.** PO has first-order welfare value for young and constrained households and can improve some dimensions of financial stability even while increasing average household borrowing.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that a realistic for-profit partial ownership contract can be embedded in a standard quantitative life-cycle housing model and can materially change housing choices.

- It shows that PO mainly operates by moving households from renting into partial ownership, not by creating a large immediate collapse in traditional homeownership.
- It also shows that PO can raise welfare substantially for the households who are most constrained by the indivisibility of housing and by mortgage regulation.

7.2 Economic intuition

- Housing is lumpy. Standard mortgage markets force households into a discrete choice: either buy a full unit and lever up heavily, or keep renting and miss the benefits of ownership.
- PO relaxes that discontinuity. A household can purchase a smaller initial ownership stake, obtain much of the utility from ownership, and then scale up later as wealth grows.
- This same mechanism explains both the welfare and stability results. PO creates more borrowers, but many of them borrow less than they would under full ownership, so the tail of risky leverage becomes thinner.

7.3 Takeaway

This paper is best understood as a quantitative analysis of how a new housing-finance contract changes the life-cycle decision to rent versus own. Its main message is that partial ownership can meaningfully improve housing affordability without simply replicating the risks of traditional homeownership. The mechanism is straightforward but important: PO substitutes a gradual ownership ladder for a large fixed leap into housing. That raises welfare for constrained households, compresses extreme leverage, and reduces default risk most strongly when house prices fall. For work on housing affordability, mortgage design, homeownership transitions, or financial stability in household balance sheets, this paper is a strong benchmark.